

Final Insights & Recommendations

ML Engine — End-to-End · Generated: 2026-05-21 06:10

1. Dataset Summary

Item	Value
File	hr_attrition_clean.csv
Shape	1,470 rows × 35 cols
Target	MonthlyIncome

2. EDA Insights (Stage 1)

INSIGHTS & RECOMMENDATIONS REPORT

Dataset : hr_attrition_clean.csv

Target : Attrition_flag

Shape : 1,470 rows × 35 columns

1. CORRELATION FINDINGS

Variables most correlated with 'Attrition_flag':

- OverTime_flag $r = +0.2461$ (positive)
- TotalWorkingYears $r = -0.1711$ (negative)
- JobLevel $r = -0.1691$ (negative)
- YearsInCurrentRole $r = -0.1605$ (negative)
- MonthlyIncome $r = -0.1598$ (negative)

KEY INSIGHT: 'OverTime_flag' shows the strongest correlation ($r = +0.2461$) with Attrition flag.

2. DATA QUALITY ASSESSMENT

Total missing cells : 0

■ No missing values detected.

3. OUTLIER SUMMARY (IQR method, 1.5×)

[illegible]

- Age 0 outliers (0.0%) [CLEAN]
- DailyRate 0 outliers (0.0%) [CLEAN]
- DistanceFromHome 0 outliers (0.0%) [CLEAN]
- Education 0 outliers (0.0%) [CLEAN]
- EmployeeNumber 0 outliers (0.0%) [CLEAN]
- EnvironmentSatisfaction 0 outliers (0.0%) [CLEAN]
- HourlyRate 0 outliers (0.0%) [CLEAN]
- JobInvolvement 0 outliers (0.0%) [CLEAN]

- JobLevel 0 outliers (0.0%) [CLEAN]
- JobSatisfaction 0 outliers (0.0%) [CLEAN]

4. SKEWNESS ASSESSMENT



Highly skewed variables ($|skew| > 1$):

- YearsSinceLastPromotion skewness = 1.984
- PerformanceRating skewness = 1.922
- Attrition_flag skewness = 1.844
- YearsAtCompany skewness = 1.765
- MonthlyIncome skewness = 1.370
- TotalWorkingYears skewness = 1.117
- NumCompaniesWorked skewness = 1.026
- JobLevel skewness = 1.025

RECOMMENDATION: Apply log/sqrt transformation to these variables before modelling.

5. MODELLING RECOMMENDATIONS



- Use the 0 variables with $corr \geq 0.3$ as features.
- Run VIF check (Tab 7) to detect multicollinearity before Regression.
- Apply IQR cleaning (Tab 3) to remove or cap extreme outliers.
- Handle any missing values (Tab 6) before splitting train/test.

3. Regression Model Results (Stage 2)

Model	R ²	MAE	RMSE	CV R ²
Random Forest	0.9081	1,050	1,417	0.9197±0.0190
Gradient Boosting	0.9002	1,097	1,477	0.9139±0.0206
Ridge Regression	0.8913	1,197	1,542	0.9024±0.0203
Lasso Regression	0.8912	1,198	1,542	0.9024±0.0203
Linear Regression	0.8912	1,198	1,542	0.9024±0.0203
Decision Tree	0.8748	1,206	1,654	0.8914±0.0178

■ Best Regression Model: **Random Forest** R² = 0.9081

4. Classification Model Results (Stage 2)

Model	Accuracy	F1	Precision	Recall	CV Acc
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K-Nearest Neighbors	84.01%	0.7980	0.7979	0.8401	82.9%
Gradient Boosting	83.67%	0.7996	0.7943	0.8367	82.7%
Random Forest	83.33%	0.7850	0.7769	0.8333	84.0%
SVM	70.07%	0.7294	0.7731	0.7007	74.1%
Decision Tree	67.01%	0.7043	0.7565	0.6701	73.6%
Logistic Regression	64.63%	0.6919	0.7916	0.6463	66.3%

■ Best Classification Model: **K-Nearest Neighbors** Accuracy = 84.01%

5. Final Recommendations

- Verify missing value imputation (Tab 6)
- Check VIF multicollinearity (Tab 7)
- Review outlier treatment (Tab 3)
- Use Random Forest for regression tasks
- Use K-Nearest Neighbors for classification tasks
- Run Predict New Data (Tab 12) for inference